

Milestone 3 Report

Retrieval-Augmented Generation (RAG), FAISS Retrieval and Cross-Encoder Reranking

Objective

The objective of this milestone was to understand the complete Retrieval-Augmented Generation pipeline. The experiments demonstrated how external knowledge retrieval improves language model performance and how incorrect retrieval can negatively affect predictions.

1. Knowledge Base Creation

A knowledge base was constructed using the correct answer text from every training example.

For every dataset row:

- The correct answer option was identified.
- The corresponding text was extracted.
- The extracted answers were stored as documents.

This collection represented the external memory source for the RAG system.

2. Dense Vector Index Creation

The embedding model:

```
sentence-transformers/all-MiniLM-L6-v2
```

was used to convert knowledge documents into dense vectors.

Each document became a numerical representation capturing semantic meaning.

A FAISS index was created using:

```
IndexFlatL2
```

FAISS enabled fast nearest-neighbour search over the vector database.

3. Zero-shot Baseline

Before applying retrieval, the model:

```
facebook/bart-large-mnli
```

was evaluated directly on questions.

This represented the baseline system where the model depended only on its pretrained knowledge.

4. FAISS Retrieval Stage

The query prompt was converted into an embedding vector.

FAISS searched the knowledge database and returned the most similar documents.

This demonstrated the first stage of RAG:

```
Query → Embedding → Vector Search → Retrieved Context
```

FAISS uses a bi-encoder approach where documents and queries are encoded independently.

Advantages:

- Very fast retrieval
- Handles large databases

Limitation:

- May miss deeper semantic relationships
-

5. Cross-Encoder Reranking

To improve retrieval quality, a Cross-Encoder model was introduced:

```
cross-encoder/ms-marco-MiniLM-L-6-v2
```

Unlike FAISS retrieval, Cross-Encoders process:

```
(question, document)
```

pairs together.

The attention mechanism directly compares query and document tokens.

Pipeline:

1. Retrieve top documents using FAISS
2. Score each document using Cross-Encoder
3. Select the highest scoring context

This improves retrieval accuracy.

6. Context Window Analysis

Experiments were performed by retrieving multiple documents and inserting them into prompts.

The generated RAG format was:

```
Context: retrieved_information Question: user_question
```

Tokenization analysis showed how increasing retrieved chunks increases context size.

This demonstrates why context-window management is important in LLM systems.

7. Retrieval-Augmented Classification

The retrieved document was added before the question.

The zero-shot classifier received:

```
External Knowledge + Question
```

instead of only:

Question

Correct context improved the model's confidence because the answer information was directly available.

8. Adversarial RAG Experiment

Incorrect context was intentionally injected into the prompt.

The experiment demonstrated:

"Garbage In, Garbage Out"

If retrieval provides wrong information, the model may trust incorrect context and reduce prediction accuracy.

This highlights the importance of retrieval quality.

9. Retrieval Evaluation

Retrieval performance was measured using Hit Rate.

A hit occurred when the correct answer appeared inside the retrieved top-k documents.

Metric:

$\text{Hit Rate} = \text{Successful Retrievals} / \text{Total Queries}$

This measured how often FAISS successfully retrieved useful information.

10. Complete RAG Pipeline

The final system combined:

Step 1: Retrieval

FAISS retrieved top-k documents.

Step 2: Reranking

Cross-Encoder selected the most relevant document.

Step 3: Augmentation

The selected context was attached to the prompt.

Step 4: Prediction

Zero-shot classifier predicted answer probabilities.

Step 5: Evaluation

Predictions were evaluated using:

MAP@3

Conclusion

This milestone demonstrated a complete modern RAG workflow:

- Vector database construction
- Dense semantic retrieval
- FAISS indexing
- Cross-Encoder reranking
- Context augmentation
- Retrieval evaluation

The experiments showed that high-quality retrieval significantly improves language model reliability, while incorrect retrieval can strongly harm performance.